

Rodrigo F. Grande Varg

Quantitative Economist — Data Science, Time Series & Forecasting

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Summary

Quantitative economist with **14+ years at the Central Reserve Bank of Peru (BCRP)**, building **forecasting, nowcasting and machine learning models** that feed monetary policy decisions every week. My work sits where statistical rigor meets production data: **high-frequency microdata panels**, automated collection pipelines, mixed-frequency models and formal out-of-sample evaluation. Delivered a **15% RMSE reduction** in the inflation nowcast and a daily credit indicator published in a **peer-reviewed journal**. **MSc in Economics and Finance** (Barcelona School of Economics) and **MSc in Applied Mathematics** (PUCP, ranked first). I teach **Machine Learning in the MSc in Artificial Intelligence at UPC**.

Professional Experience

Central Reserve Bank of Peru (BCRP)

Sep 2019 – Present

Senior Price Specialist

- Built **mixed-frequency and machine learning nowcasting models** for inflation (XGBoost, forecast combination), achieving a **15% RMSE reduction** over the previous specification.
- Own the **short-term inflation forecast: two weekly forecast reports** (headline and core inflation) used as direct input for monetary policy decisions.
- Developed a **daily high-frequency credit indicator** (Random Forest and dynamic network connectedness) to anticipate turning points in the credit cycle; published in the *Latin American Journal of Central Banking*.
- Build and curate **high-frequency price microdata panels**: integration of heterogeneous sources, missing-data treatment and consistency checks on daily and weekly series.
- Automated data collection with **Python web scraping**, using reproducible extraction, cleaning and validation pipelines.
- Built economic indicators from **unstructured data (NLP)** and **interactive dashboards** (Plotly Dash, Power BI) to communicate results to decision-makers.

Central Reserve Bank of Peru (BCRP)

Jul 2013 – Aug 2018

Economic Outlook Specialist

- Implemented leading indicators of economic activity (BCRP Working Paper 2017-01).
- Forecast macroeconomic variables and contributed to the *Inflation Report*.

Central Reserve Bank of Peru (BCRP)

May 2012 – Jun 2013

Balance of Payments Specialist

- Compiled and analyzed the financial account of the balance of payments.

Teaching

Universidad Peruana de Ciencias Aplicadas (UPC)

Mar 2026 – Present

Lecturer, Machine Learning — MSc in Artificial Intelligence: statistical learning, linear and regularized models, SVM, probabilistic models, clustering and advanced model evaluation; Python labs (scikit-learn) on reproducible modeling pipelines, cross-validation and preventing data leakage.

Universidad San Martín de Porres

Mar 2020 – Jul 2024

Lecturer, Applied Time Series and Economic Growth (ARIMA, forecasting and model evaluation in Python).

Universidad Nacional de Ingeniería

Mar 2018 – Jul 2018

Lecturer, International Economics.

Education

Barcelona School of Economics (Universitat Pompeu Fabra)

2018 – 2019

MSc in Economics and Finance, Finance Program — GPA 8.01/10

Relevant courses: Econometric Methods, Financial Econometrics, Machine Learning for Finance, Risk Management in Banks.

Selected Projects (github.com/Rodgrandez)

- **peru-inflation-nowcasting:** nowcasting Lima CPI inflation with U-MIDAS, Almon-MIDAS, Ridge/LASSO, XGBoost and forecast combination on daily BCRP data; expanding-window evaluation 2015–2026 and Diebold-Mariano tests.
- **lima-food-prices-pipeline:** automated daily pipeline for Lima wholesale food prices (MIDAGRI-SISAP, 2010–2026) with published data-quality checks, price-pressure indicators and a live dashboard.
- **credit-risk-pd-validation:** PD scorecard (WoE logistic), XGBoost challenger with SHAP, LGD and expected loss, plus a second-line validation report (discrimination, calibration, PSI, backtesting).
- **bank-account-fraud-detection:** fraud detection on 1M bank account applications (Feedzai BAF) with strictly time-based validation, cost- and budget-based alert decisions, calibration, a fairness audit by age, drift monitoring and a model card.

Research and Publications

- Grandez, R., & Pardo-Figueroa, R. (2026). *A high frequency indicator of credit in Peru: A Random Forests and dynamic network connectedness approach.* Latin American Journal of Central Banking. **(Peer-reviewed)**
- Grandez, R., & Pardo, R. (2025). *Nowcasting and Backcasting Credit with Mixed-Frequency Data and Machine Learning: Evidence from Peru.* SSRN Working Paper No. 5408526.
- Flores, J., & Grandez, R. (2024). Machine learning methods to forecast inflation in Peru. *Revista Moneda*, No. 200, BCRP (in Spanish).
- Pérez Forero, F., Ghurra Aguilar, O., & Grandez Vargas, R. (2017). *A Leading Indicator for the Peruvian Economic Real Activity.* BCRP Working Paper No. 2017-01.
- MSc theses: *SABR: A stochastic volatility model in practice* (BSE, awarded) and *Option pricing with Lévy processes* (PUCP).

Technical Skills

Machine learning: Supervised and unsupervised learning, XGBoost, Random Forest, SVM, regularization, clustering, applied NLP; scikit-learn, PyTorch.

Statistics and model evaluation: Out-of-sample evaluation, cross-validation, data-leakage prevention, forecast combination, error metrics (RMSE, MAE), Diebold-Mariano tests.

Time series and econometrics: ARIMA/SARIMA, VAR/VECM, cointegration (Engle-Granger, Johansen), unit-root tests (ADF, PP, KPSS), mixed-frequency models (MIDAS/U-MIDAS), HP/BK filters.

Data engineering: Python (NumPy, Pandas), SQL, MongoDB, Git; microdata panel construction, web scraping, data quality control and reproducible pipelines.

Other tools: R, MATLAB, EViews; Plotly Dash, Power BI, Tableau.

Languages: Spanish (native), English (advanced), Portuguese (basic).